Hyun Jeong (鄭　玄)

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Personal Statement

I am a Ph.D. student studying gravity both from the phenomenological and theoretical perspectives. My research began with inflationary cosmology, with a particular focus on its observational consistency, while deepening my understanding of the underlying theoretical challenges. Recently, I have been exploring scattering amplitudes and their applications to black hole physics, and I look forward to advancing this direction through discussions with you all.

Education

Apr 2024 – present	**Ph.D. in Physics (in progress), The University of Tokyo** Advisor: Prof. Jun'ichi Yokoyama
Apr 2022 – Mar 2024	**M.Sc. in Physics, The University of Tokyo** Advisor: Prof. Jun'ichi Yokoyama Thesis: "Cosmological Evolution after R^2 inflation"
Apr 2018 – Mar 2022	**B.Sc. in Physics, The University of Tokyo**

Research Experience

Feb 2026 – Mar 2026	**Visiting Researcher, National Taiwan University** Host: Prof. Yu-Tin Huang
Jan 2025 – Mar 2025	**Visiting Researcher, YITP, Kyoto University** Host: Dr. Katsuki Aoki

Grants and Fellowship

Apr 2024 – Mar 2027	**JSPS Research Fellowship for Young Scientists (DC1)**, Japan Society for the Promotion of Science (JSPS), Grant No. 24KJ0902.
Apr 2022 – Mar 2027	**Forefront Physics and Mathematics Program to Drive Transformation (FoPM)**, The University of Tokyo.

Languages

Japanese (native). English and Korean (fluent). German (basic).

Publications

1. K. Aoki, F-Y. Cheng, A. Cristofoli, Y-t. Huang and **H. Jeong**, An Effective S-matrix Approach to Low-Frequency Waveforms from Black Hole Mergers, [arXiv: 2609.10329].
2. D. Akpinar, K. Aoki, A. Cristofoli, **H. Jeong** and K. Yoshimura, *Black Hole Thermodynamics Meets On-Shell Amplitudes: Local Detailed Balance and Thermal Spectrum from Spin Universality and Unitarity*, [arXiv: 2606.13599].
3. K. Aoki, A. Cristofoli, **H. Jeong**, M. Sergola and K. Yoshimura, *Quantum Effects for Black Holes with On-Shell Amplitudes*, *JHEP* **12** (2025) 163.
4. **H. Jeong**, K. Kamada, A. A. Starobinsky and J. Yokoyama, *Reheating process in the R^2 inflationary model with the baryogenesis scenario*, *JCAP* **11** (2023) 023.

Presentations (International Conferences)

1. **H. Jeong**, "Gravitational Memory in Black Hole Mergers from Scattering Amplitudes", Theoretical Developments in Gravitational Wave Physics, Kyoto University, Jun 15-19, 2026 (oral).
2. **H. Jeong**, "Quantum Effects for Black Holes with On-Shell Amplitudes", Nagoya workshop on General Relativity, Nagoya University, Jan 26-28, 2026 (oral).
3. **H. Jeong**, "Quantum Effects for Black Holes with On-Shell Amplitudes", The 34th Workshop on General Relativity and Gravitation in Japan (JGRG 34), Kyoto University, Jan 19-23, 2026 (oral).
4. **H. Jeong**, "The role of Higgs field after R^2 inflation", COSMO'24, Kyoto University, Oct 21-25, 2024 (poster).
5. **H. Jeong**, "Are primordial black holes produced after R^2 inflation?", The 32nd Workshop on General Relativity and Gravitation in Japan (JGRG 32), Nagoya University, Nov 27-Dec 1, 2023 (oral).
6. **H. Jeong**, "R^2 inflation with the baryogenesis scenario", FoPM International Symposium, The University of Tokyo, Feb 6-8, 2023 (poster).

Presentations (Domestic Conferences)

1. **H. Jeong**, "Gravitational Memory in Black Hole Mergers from Scattering Amplitudes", 81th Annual Meeting of The Physical Society of Japan, The University of Tokyo, Sep 14-17, 2026 (oral).
2. **H. Jeong**, "Effects of Hawking Radiation on Black Hole Scattering", 80th Annual Meeting of The Physical Society of Japan, Hiroshima University, Sep 16-19, 2025 (oral).
3. **H. Jeong**, "The role of Higgs field after R^2 inflation", Joint workshop on General Relativity and Cosmology 2025, Nishiyatsu-Onsen, Mar 10-11, 2025 (oral).
4. **H. Jeong**, "The role of Higgs field after R^2 inflation", 79th Annual Meeting of The Physical Society of Japan, Hokkaido University, Sep 16-19, 2024 (oral).
5. **H. Jeong**, "Are primordial black holes produced after R^2 inflation?", 2024 Spring Meeting of The Physical Society of Japan, online, Mar 18-21, 2024 (oral).
6. **H. Jeong**, "Toward a theoretically and phenomenologically consistent model", Joint workshop on General Relativity and Cosmology 2024, Sunshine Shirako, Mar 6-8, 2024 (oral).
7. **H. Jeong**, "Reheating process in the R^2 inflationary model with the baryogenesis scenario", 78th Annual Meeting of The Physical Society of Japan, Tohoku University, Sep 16-19, 2023 (oral).
8. **H. Jeong**, "R^2 inflation with the baryogenesis scenario", Joint workshop on General Relativity and Cosmology 2023, Okutama Town Social Welfare Council, Mar 13-14, 2023 (oral).
9. **H. Jeong**, "R^2 inflation with the baryogenesis scenario", The 35th Rironkon Symposium, Corasse Fukushima, Dec 21-23, 2022 (poster).

Seminars

1. **H. Jeong**, "An Effective S-matrix Approach to Black Hole Mergers", Niels Bohr Institute, University of Copenhagen, Oct 5, 2026.
2. **H. Jeong**, "An Effective S-matrix Approach to Black Hole Mergers", Higgs Centre for Theoretical Physics, The University of Edinburgh, Oct 2, 2026.
3. **H. Jeong**, "An Effective S-matrix Approach to Black Hole Mergers", CTP Seminar, Queen Mary University of London, Sep 24, 2026.
4. **H. Jeong**, "Quantum Effects for Black Holes with On-Shell Amplitudes" (blackboard presentation-style), NTU Strings Seminar, National Taiwan University, Mar 18, 2026.
5. **H. Jeong**, "Quantum Effects for Black Holes with On-Shell Amplitudes" (slide presentation-style), NTU Strings

Seminar, National Taiwan University, Feb 6, 2026.

Awards

1. **H. Jeong,** Outstanding Performance Award, Final Examination, Forefront Physics and Mathematics Program to Drive Transformation (FoPM), The University of Tokyo.
2. **H. Jeong**, Outstanding Presentation Award Gold Prize (out of 112), The 34th Workshop on General Relativity and Gravitation in Japan (JGRG 34), Kyoto University, Jan 19-23, 2026.
3. **H. Jeong**, Outstanding Presentation Award Gold Prize (out of 14), Joint workshop on General Relativity and Cosmology 2023, Okutama Town Social Welfare Council, Mar 13-14, 2023.
4. **H. Jeong**, Oral Award (2nd place out of 39), Gravity & Cosmology Section, "Starobinsky Model vs. Higgs Inflation Model", The 52nd Astronomy & Astrophysics Summer School for Young Researchers, Aug 23-26, 2022.
5. Special Prize: Japanese Institute of Certified Public Accountants Award, Mathematics Koshien 2017 – As a Team Leader of Toin Gakuen Secondary Education School (Pre-university).

Outreach and Writing

1. Student essay, "Black Holes as an Energy Source", FoPM graduate course "Scientific Writing, Publication, and Communication" (FY2022).
2. "Singularity Theorems and Future Prospects", Physics Lab. (Dept. of Physics) Advent Calendar 2020, Day 22 (in Japanese).
3. "Convergence Factors from the Perspective of Time Ordering", Physics Lab. (Dept. of Physics) Advent Calendar 2020, Day 19 (in Japanese).

Referees

Prof. Jun'ichi Yokoyama | The University of Tokyo | junichi.yokoyama@ipmu.jp

Prof. Kohei Kamada | Hangzhou Institute for Advanced Study, UCAS | kohei.kamada@ucas.ac.cn

Dr. Katsuki Aoki | Kyoto University | katsuki.aoki@yukawa.kyoto-u.ac.jp